Data Analytics Project – End-to-End Data Analysis & Visualization

📌 Overview

This project demonstrates a complete data analytics workflow, from raw data ingestion to interactive dashboard creation and business insights generation. It combines Python-based data analysis, SQL querying, and Power BI visualization to deliver actionable insights and a professional reporting output suitable for business decision-making.

Dataset

- **Source:** (Add dataset source or mention "Kaggle / Internal / Open dataset")
- **Format:** CSV / Excel / SQL Database
- **Description:** The dataset contains structured records used for analysis, including key business-related attributes such as categories, timestamps, numerical metrics, and categorical variables.

🛠 Tools & Technologies

- **Python** (Pandas, NumPy, Matplotlib, Seaborn)
- **SQL** (PostgreSQL / MySQL / SQL Server)
- **Power BI** (Interactive Dashboard Creation)
- **Jupyter Notebook** (EDA & Data Cleaning)

🔄 Project Workflow (Steps)

1. Data Loading

- Imported dataset using Python (Pandas)
- Performed initial inspection (head(), info(), describe())

2. Exploratory Data Analysis (EDA)

- Identified trends, patterns, and anomalies
- Generated visualizations (bar charts, histograms, heatmaps)
- Analyzed correlations between variables

3. Data Cleaning & Preprocessing

- Handled missing values (imputation or removal)
- Removed duplicates and irrelevant columns
- Standardized data formats and column names
- Detected and treated outliers

4. SQL Analysis

- Loaded cleaned dataset into SQL database (PostgreSQL/MySQL/SQL Server)
- Executed queries for:
  - Aggregations (SUM, AVG, COUNT)
  - Grouping and filtering data
  - Business-specific insights (KPIs, trends)
- Joined tables where applicable

5. Power BI Dashboard

- Imported cleaned dataset into Power BI
- Built interactive visualizations:
  - KPI cards
  - Trend charts
  - Category-wise breakdowns
  - Slicers and filters for interactivity
- Designed a clean and professional dashboard layout

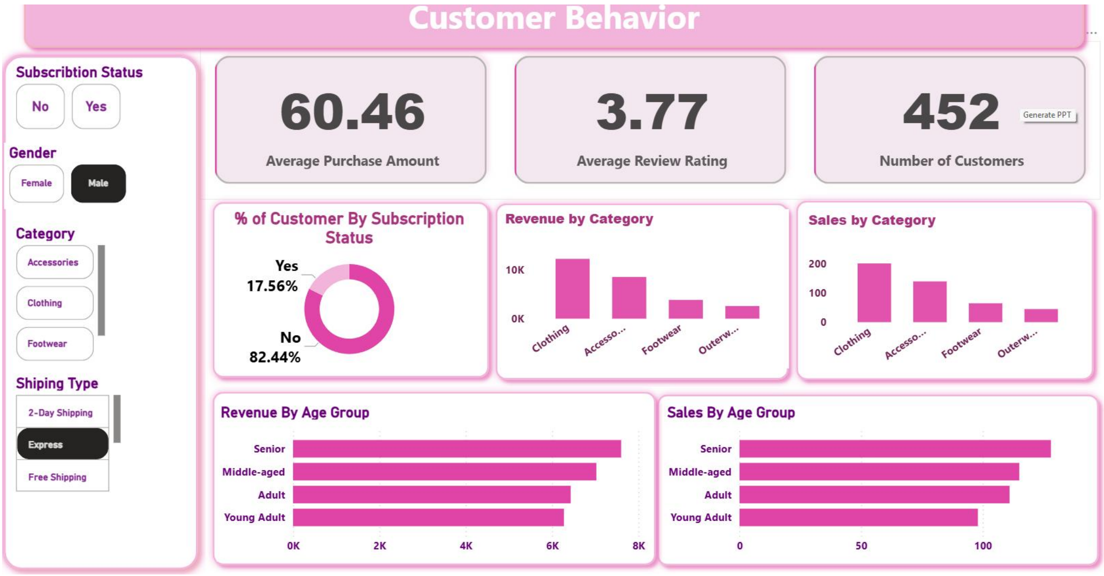
6. Reporting & Presentation

- Summarized findings in a structured report
- Created a presentation using **Gamma**
- Highlighted key insights and business recommendations

📊 Dashboard

The Power BI dashboard includes:

- Key Performance Indicators (KPIs)
- Time-series trend analysis
- Category-wise and regional breakdowns
- Interactive filters and slicers
- Drill-down capabilities for deeper insights

- 

🔍 Key Results / Insights

- Identified major trends influencing business performance
- Discovered top-performing categories/products/segments
- Highlighted data quality issues impacting analysis
- Found correlations between key business variables
- Provided actionable recommendations for decision-making